

Representing Integers on the Number Line

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A **number line** is a straight line used to represent numbers in order.

The **origin** is 0, which separates **positive integers** and **negative integers**.

Positive integers are located to the **right** of 0.

Negative integers are located to the **left** of 0.

The farther a number is to the right, the **greater**
its value.

The farther a number is to the left, the **smaller** its value.

Each integer has a unique position on the number line.

Locating integers on the number line helps compare numbers and understand their order.

Examples

Example 1

Locate the integers -4 , 0 , and 5 on a number line from -6 to 6 .

The integer -4 is located to the left of 0 .

The integer 0 is the origin.

The integer 5 is located to the right of 0 .

Example 2

Which integer is located 4 units to the left of 0?

Moving 4 units to the left of 0 gives:

$$\boxed{-4}$$

Example 3

Which integer is located 6 units to the right of 0?

Moving 6 units to the right of 0 gives:

$\boxed{6}$

Example 4

What integer is halfway between -8 and 0 on the number line?

The integer halfway between -8 and 0 is:

$$\boxed{-4}$$

Example 5

List the integers from left to right:

$3, -5, 0, -1, 4$

From left to right:

$-5, -1, 0, 3, 4$

Practice Exercises

Practice 1

Draw a number line from -6 to 6 . Plot the integers -4 , 0 , and 5 .

Practice 2

Draw a number line from -8 to 8 . Plot the integers -7 , -2 , and 4 .

Practice 3

Draw a number line from -10 to 10 . Plot the integers -9 , 1 , and 8 .

Practice 4

Draw a number line from -5 to 5 . Plot the integers -3 , 2 , and 5 .

Practice 5

Draw a number line from -12 to 12 . Plot the integers -10 , 0 , and 9 .

Practice 6

Which integer is located 4 units to the left of 0?

Practice 7

Which integer is located 6 units to the right of 0?

Practice 8

What integer is halfway between -8 and 0 on the number line?

Practice 9

List the integers from left to right:

$3, -5, 0, -1, 4$

Practice 10

List the integers from left to right:

$-7, 2, -3, 5, 0$

Activities

Activity 1

Draw a number line from -7 to 7 . Plot the integers -5 , 0 , and 6 .

Activity 2

Draw a number line from -9 to 9 . Plot the integers -8 , -1 , and 7 .

Activity 3

Draw a number line from -11 to 11 . Plot the integers -10 , 2 , and 9 .

Activity 4

Draw a number line from -6 to 6 . Plot the integers -4 , 3 , and 6 .

Activity 5

Draw a number line from -13 to 13 . Plot the integers -11 , 0 , and 10 .

Activity 6

Which integer is located 5 units to the left of 0?

Activity 7

Which integer is located 7 units to the right of 0?

Activity 8

What integer is halfway between -10 and 0 on the number line?

Activity 9

List the integers from left to right:

$2, -6, 1, -2, 5$

Activity 10

List the integers from left to right:

$-8, 3, -4, 6, 0$

Thank you!